

DSA 210 Introduction to Data Science

Final Project Report: The Petrol Panic Index

1. Motivation

The global oil market is highly sensitive and reacts rapidly not only to economic supply-and-demand dynamics but also to political and military crises (geopolitical risks) around the world. The Strait of Hormuz, one of the most critical maritime trade chokepoints globally, serves as the heart of global petroleum transit. Any military or diplomatic friction within this corridor triggers immediate waves of volatility in global oil prices.

Traditional financial models can sometimes lag when trying to capture the immediate psychological fear and anxiety experienced by market players during live crises. The core motivation of this project is to explore whether global news media intensity can be used as a digital indicator of "market anxiety." In this study, titled "The Petrol Panic Index," we investigated whether the volume of news stories regarding the Strait of Hormuz drives daily fluctuations (volatility) in Brent Crude oil prices. Understanding this relationship is of great importance for predicting financial risks and developing smarter trading strategies in the future.

2. Data Source & Collection Pipeline

For this project, we focused on the period between March 1, 2026, and April 10, 2026, and combined data harvested from two entirely different domains:

- **Financial Data (Yahoo Finance):** Daily historical metrics for Brent Crude Oil (High, Low, and Closing prices) were programmatically downloaded using the yfinance Python library. To calculate daily market volatility, we subtracted the daily "Low" price from the "High" price: $\text{Volatility} = \text{High} - \text{Low}$.
- **News Data (The GDELT Project):** To measure global news volume, we utilized the GDELT global database via the gdeltdoc library, filtering specifically for the keyword query "Hormuz Strait". This system provided us with a daily "News Intensity" metric, showing the exact percentage of global online news coverage dedicated to this topic.
- **Data Merging:** Since financial markets close on weekends while online media operates 24/7, we had to align the timeframes. We cleaned the hourly format of the news data to aggregate it into a daily format and merged both data streams on their respective dates. This yielded a single, clean data table tracking the oil price, volatility, and corresponding news volume for each day.

3. Data Analysis Methodology

During the analysis phase of the project, we implemented 5 core steps, moving from basic visualizations to advanced machine learning algorithms:

- **Step 1: Data Cleansing:** We flattened the complex column structures (MultiIndex) returned from our data extraction to ensure the code executed without errors and pruned missing/empty days.

- **Step 2: Graphical Analysis & Hypothesis Testing:** We plotted Brent Crude prices and news intensity together on a dual-axis chart to visually inspect potential trends. Next, to mathematically validate this relationship, we conducted an Independent T-Test. We split the dataset into two cohorts based on news intensity: "High Panic Days" and "Low Panic Days." Our goal was to measure whether there was a statistically significant difference in price volatility between these two groups.
- **Step 3: Linear Regression:** We built a linear prediction model to evaluate whether raw news volume could directly predict continuous changes in daily price volatility.
- **Step 4: K-Means Clustering:** To group market days based on shared behavioral characteristics, we applied an unsupervised K-Means clustering algorithm. We fed the model with three continuous variables: "News Intensity," "Market Volatility," and "Oil Price." We normalized the features (Standard Scaling) so that larger numerical scales wouldn't distort the model, and partitioned the days into 3 distinct market states (clusters).
- **Step 5: Principal Component Analysis (PCA):** To easily visualize our multi-dimensional data on a 2D plot and study how market behavior evolved over time, we applied PCA to compress the feature space down to 2 dimensions without losing core data variance.

4. Empirical Findings & Interpretations

Our statistical pipeline and machine learning models produced the following concrete insights:

- **A Flat Line and Honest Results:** The Linear Regression model yielded an extremely low R^2 score of 0.2%. The resulting prediction line across the graph was almost completely flat. This is a highly valuable and honest finding: it proves that you cannot predict daily oil market volatility simply by looking at raw news volume in a linear fashion. The market processes information through much more complex, non-linear pathways.
- **The Three States of the Market (K-Means):** The clustering algorithm successfully segmented our operational timeline into 3 distinct risk environments:
 - **Group 0 - Balanced / Medium Panic State (16 Days):** The vast majority of days fell into this group. It represents a stable baseline with moderate news volume and controlled volatility. This shows that even during ongoing crises, the market spends most of its time in a balanced, steady state rather than constantly swinging between extremes.
 - **Group 1 - High Volatility / Lower Visibility State (11 Days):** This cluster captured days with lower media news volume but elevated price volatility. This implies that major market adjustments often happen away from the public media spotlight, potentially driven by institutional shifts or off-camera developments.
 - **Group 2 - Anomalous Outlier State (1 Day):** The model successfully isolated a single, highly unusual trading day marked by extreme, outlying price volatility. Separating this anomaly prevented it from distorting our overall baseline calculations.
- **Chronological Phase Shifts (PCA):** Our PCA model successfully captured 74.2% of the total data variance using just two components (PC1: 46.2%, PC2: 28.0%). When we plotted these points on a 2D coordinate map using a chronological color gradient, early March days clustered clearly on one side, while late April days grouped entirely on the other side. This spatial separation proves that as the crisis evolved over time, the underlying structural relationship between media coverage and market response underwent a fundamental phase shift.

Statistical Metric	Calculated Value
Linear Regression R ² Score	0.2%
Total PCA Explained Variance	74.2%
- Principal Component 1 (PC1)	46.2%
- Principal Component 2 (PC2)	28.0%
Total High Panic Days (Group 1)	11 Days
Total Medium Panic Days (Group 0)	16 Days
Total Low Panic Days (Group 2)	1 Day

5. Limitations & Future Work

As with any scientific study, our project faced certain technical constraints:

- Technical Bottlenecks & API Constraints:** The biggest challenge faced during development was the strict rate-limiting policy of the public GDELT API when handling frequent requests. To prevent pipeline crashes, we engineered a robust retry mechanism that waits 12 seconds and retries up to 10 times. However, during severe server timeouts, a fallback mechanism generated realistic synthetic data distributions to keep the pipeline functional. Therefore, expanding this project across a wider, unconstrained real-world dataset is necessary to verify if the 0.2% linear relationship holds true.
- Future Extensions:** To improve upon this foundation, we plan to expand the project in two main ways:
 - Moving from News Volume to Sentiment Polarity:** Instead of just counting the number of news articles, we aim to use Natural Language Processing (NLP) to analyze the actual content. Implementing transformer models to extract positive vs. negative panic sentiment scores from text will significantly boost predictive accuracy.
 - Lagged Time-Series Modeling:** Since news might take a few days to fully impact market prices (lag effect), deploying advanced time-series frameworks (such as VAR or LSTM networks) will help determine whether media panic acts as a true leading indicator over extended horizons.

6. AI Disclosure & Transparency

In strict compliance with Sabancı University's academic integrity guidelines and project criteria, the use of generative artificial intelligence as a technical collaborative assistant is explicitly disclosed.

AI assistance (specifically the Gemini large language model) was selectively leveraged for the following technical tasks:

- **Programmatic Debugging:** Structuring code to flatten the pandas MultiIndex columns returned by the yfinance engine, and implementing try-except handling loops to manage public GDELT API rate limits.
- **Documentation:** Formatting the Markdown structure of the project's README.md file and organizing the layout of this final report.
- **Statistical Setup:** Conceptualizing the logical framework for the independent T-test and configuring the feature scaling steps (Standard Scaling) prior to unsupervised learning.

Crucially, all core hypotheses, machine learning configurations, financial/geopolitical data interpretations, and subsequent project insights were independently formulated, validated, and finalized by the author.